

Radial extension of an isometry between unit spheres

MathlibAnnex.Sphere.radialExtension

def

Extends a bijective sphere isometry to all vectors by retaining each radius and transporting its unit direction.

Statement

Let X, Y be real normed spaces, let $S_X = \{x : \|x\| = 1\}$ and $S_Y = \{y : \|y\| = 1\}$ carry their ambient chord distances, and let $\Delta : S_X \rightarrow S_Y$ be a bijective isometry. The radial extension $R_\Delta : X \rightarrow Y$ sends zero to zero and sends a nonzero vector to its radius times the image of its unit direction.

Definition

For $x \neq 0$, put $\hat{x} = \|x\|^{-1}x$. Since $\|x\| > 0$,

$$\|\hat{x}\| = \|x\|^{-1}\|x\| = 1.$$

Thus $\hat{x} \in S_X$, so the formula

$$R_\Delta(x) = \begin{cases} 0, & x = 0, \\ \|x\|\Delta(\hat{x}), & x \neq 0 \end{cases}$$

is well-defined. Normalization fixes unit vectors and is unchanged by positive rescaling, as the cited normalization lemmas state.

Assumptions

Both spaces are real normed spaces and Δ is an isometry equivalence of their unit spheres. Neither space is assumed finite-dimensional, complete or nonzero. In the zero space there is no nonzero vector requiring normalization.

Conclusion

This defines R_Δ on all of X . The separately cited norm theorem gives $\|R_\Delta(x)\| = \|x\|$; for $x \neq 0$ this follows from $\|\Delta(x/\|x\|)\| = 1$, and zero is immediate. The sphere-agreement theorem gives $R_\Delta(u) = \Delta(u)$ for $u \in S_X$.

This is a radial, positively homogeneous extension. Its defining operation uses positive radii; the declaration does not provide a linear extension theorem. In the classification argument R_Δ is the map used for analytic and topological comparisons; a linear isometry supplied at a later stage is a separate object.

Main citations

- Exact declaration and its source — `MathlibAnnex.Sphere.radialExtension`
- Normalizing a nonzero vector into the sphere — `MathlibAnnex.Sphere.normalizeToSphere`
- Normalization fixes a unit vector — `MathlibAnnex.Sphere.normalizeToSphere_unit`
- Positive rescaling preserves the unit direction — `MathlibAnnex.Sphere.normalizeToSphere_pos_smul`
- The extension preserves radii — `MathlibAnnex.Sphere.radialExtension_norm`
- The extension agrees on the sphere — `MathlibAnnex.Sphere.radialExtension_on_sphere`

Lean source signature (exact)

```
def radialExtension
  (e : Metric.sphere (0 : X) 1 ≈i Metric.sphere (0 : Y) 1) (x : X) : Y := by
  classical
  exact if hx : x = 0 then 0 else
    ‖x‖ • (e (normalizeToSphere x hx) : Y)
```

In the source	Mathematical meaning
<code>(e : Metric.sphere (0 : X) 1 ≈_i Metric.sphere (0 : Y) 1)</code>	The same bijective chord-distance isometry $\Delta : S_X \rightarrow S_Y$, where $S_X = \{x : \ x\ _X = 1\}$ and $S_Y = \{y : \ y\ _Y = 1\}$. The surrounding binders make X, Y real normed spaces.
<code>(x : X) : Y</code>	Input $x \in X$ and output vector $R_\Delta(x) \in Y$.
<code>if hx : x = 0 then 0 else</code>	The zero branch gives $R_\Delta(0) = 0$. In the other branch hx supplies $x \neq 0$.
<code>normalizeToSphere x hx</code>	The sphere point with underlying vector $\hat{x} = x/\ x\ _X$; nonzeroness makes its norm equal to one.

In the source	Mathematical meaning
<code>(e (normalizeToSphere x hx) : Y)</code>	The vector $\Delta(\hat{x}) \in Y$. The cast forgets sphere membership evidence.
<code>‖x‖ • (e (normalizeToSphere x hx) : Y)</code>	The complete nonzero branch $R_\Delta(x) = \ x\ _X \Delta(\hat{x})$. Together with the zero branch this is the entire defining formula.

▼ Exact surrounding binder context (separate excerpt)

```
namespace MathlibAnnex
namespace Sphere

universe u v

variable {X : Type u} {Y : Type v}
```

[NormedAddCommGroup X] [NormedSpace $\mathbb{R}$ X]
[NormedAddCommGroup Y] [NormedSpace $\mathbb{R}$ Y]

▼ **Exact content identity**

Declaration: MathlibAnnex.Sphere.radialExtension

Accepted content SHA-256: dadb75eadd5f641823000ab853b1c072fbdac40655c416aecea62f0dc9fadb9

Accepted source guide SHA-256: d2ccd2c6df145ab9877bfc1c80a1f9d38650e9b2a762f09c9c2e2c6bd46a1981

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73